DURGA S



Aspiring Data Analyst | AI/ML Engineer | Automation Developer

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PROFESSIONAL SUMMARY

Recent B.Tech Artificial Intelligence and Data Science graduate with hands-on experience building end-to-end machine learning applications, computer vision workflows and RPA automation solutions. Completed two internships in machine learning and Azure cloud computer vision. Author of an IEEE conference paper on explainable traffic accident hotspot forecasting using Transformers. Skilled in Python, Pandas, NumPy, OpenCV, SQL and UiPath, with a strong foundation in data preprocessing, model evaluation and business intelligence. Seeking entry-level roles in data analytics, AI/ML and process automation.

EDUCATION

B.Tech – Artificial Intelligence and Data Science *Aug 2022 – 2026*
K.S.Rangasamy College of Technology, Tiruchengode | CGPA: 8.2
Higher Secondary (HSC) – 79.64% – *Vani Vidyalaya Matric. Hr. Sec. School, Salem* *2022*
Secondary School Leaving Certificate (SSLC) – 89% – *Vani Vidyalaya Matric. Hr. Sec. School, Salem* *2020*

TECHNICAL SKILLS

Programming & Web: Python, SQL, HTML, CSS, JavaScript, PHP
Machine Learning & AI: Supervised Learning, Computer Vision, Transformers, Explainable AI, Spatio-Temporal Modeling, OCR, Data Preprocessing, Model Evaluation
Libraries & Frameworks: NumPy, Pandas, OpenCV, Transformers **Cloud & Automation:** Azure AutoML, UiPath Studio (RPA)
Database & Analytics: MySQL, Business Intelligence & Analytics, Data Analysis **IoT:** Internet of Things (NPTEL Elite)
Tools: VS Code, Git/GitHub, AutoCAD, Canva, MS Office
Soft Skills: Teamwork, Adaptability, Time Management, Decision Making, Communication

FINAL YEAR PROJECT & PUBLICATION

Transformer-Driven Spatio-Temporal Modeling for Explainable Traffic Accident Hotspot Forecasting *2025 – 2026*
Tools: Python, Pandas, NumPy, Transformer models

- Published in an IEEE conference hosted by Nepal College of Information Technology (NCIT), Nepal.
- Developed a Transformer-based spatio-temporal model to forecast traffic accident hotspots from historical accident data.
- Added explainability to identify and interpret the key factors driving each prediction, supporting data-driven road safety decisions.

INTERNSHIP EXPERIENCE

Machine Learning Intern – AVATAR Robotics and Automation *Jan 2024 – Mar 2024*

- Developed machine learning models for industrial automation use cases.
- Performed data preprocessing, feature preparation and model evaluation to improve model performance; collaborated with the team on model deployment.

Azure Cloud & Computer Vision Intern – Microsoft Girls! Go Ready Program *Nov 2024 – Jan 2025*

- Implemented Azure AutoML for OCR and number plate detection; designed scalable cloud workflows for real-time computer vision tasks.

PROJECTS

Emotional Prediction System *Python, Pandas*

- Built a machine learning model to detect emotions from text and facial expressions, improving recognition accuracy.

Tourism Management System *HTML, CSS, JavaScript, SQL, PHP*

- Developed a full-stack web application for tour planning and booking with automated recommendations.

Flight Ticket Booking Automation *UiPath Studio*

- Automated end-to-end flight booking with exception handling and a reusable bot design.

CERTIFICATIONS

- UiPath Certified Professional – Automation Developer Associate
- NPTEL – Business Intelligence & Analytics
- NPTEL – Internet of Things (Elite)
- GUVI – Practical Machine Learning

ACHIEVEMENTS & ACTIVITIES

- Published a research paper in an IEEE conference (Nepal College of Information Technology) on traffic accident hotspot forecasting.
- Presented at the National Conference: AI for Emotional Synchronization.
- Participated in TNWISE 2025 (TANCAM) – Tamil Nadu's largest hackathon for women in STEM.